

Print and check off each connection during build

COLOUR KEY			
Green = In use	Orange = Handle with care	Red = Reserved / do not use	Grey = Spare

GPIO Pin Assignments					
GPIO	Direction	Connected to	Signal	Notes	Checked
GPIO0	INPUT	Boot strapping pin	BOOT	Do not pull LOW at boot	☐
GPIO1	INPUT	Battery voltage divider	ADC	100k/33k -> 3.13V max at 12.6V	☐
GPIO2	OUTPUT	BTS7960 RPWM	PWM	Fill direction 0-255	☐
GPIO3	OUTPUT	Passive piezo buzzer	PWM	ledcWriteTone()	☐
GPIO4	INPUT	Fill flow sensor	PULSE	7.5k/15k divider 3.33V	☐
GPIO5	INPUT	Drain flow sensor	PULSE	7.5k/15k divider 3.33V	☐
GPIO6	INPUT	Tank full sensor	DIGITAL	7.5k/15k divider 3.33V	☐
GPIO7	INPUT	Tank sensor detect pin 4	DIGITAL	15k pull-up to 3.3V	☐
GPIO8	OUTPUT	BTS7960 R_EN	DIGITAL	Hold HIGH to enable	☐
GPIO9	OUTPUT	BTS7960 LPWM	PWM	Drain direction 0-255	☐
GPIO10	OUTPUT	BTS7960 L_EN	DIGITAL	Hold HIGH to enable	☐
GPIO11	OUTPUT	TFT SCL (SPI clock)	SPI	SPI2 SCK	☐
GPIO12	OUTPUT	TFT SDA (SPI MOSI)	SPI	SPI2 MOSI	☐
GPIO13	OUTPUT	TFT CS (chip select)	SPI	Active LOW	☐
GPIO14	OUTPUT	TFT DC (data/command)	DIGITAL	HIGH=data LOW=cmd	☐
GPIO15	OUTPUT	TFT RES (reset)	DIGITAL	Active LOW	☐
GPIO16	I/O	Pololu A pin via D3 1N4148	DIGITAL	Pulls A LOW + reads SW timing	☐
GPIO17	OUTPUT	Pololu OFF pin	DIGITAL	Pulse HIGH = forced shutdown	☐
GPIO18	I/O	I2C SDA (Wire0)	I2C	DS3231 + ABP2 (4.7k pull-up)	☐
GPIO19	I/O	USB D- RESERVED	USB	Keep free for USB	☐
GPIO20	I/O	USB D+ RESERVED	USB	Keep free for USB	☐
GPIO21	INPUT	Encoder DT (direction)	DIGITAL	Read after CLK interrupt	☐
GPIO26	I/O	Internal flash - do not use	-	Do not connect	☐
GPIO27	I/O	Internal flash - do not use	-	Do not connect	☐
GPIO28	I/O	Internal flash - do not use	-	Do not connect	☐
GPIO29	I/O	Internal flash - do not use	-	Do not connect	☐
GPIO30	I/O	Internal flash - do not use	-	Do not connect	☐
GPIO31	I/O	Internal flash - do not use	-	Do not connect	☐
GPIO32	I/O	Internal flash - do not use	-	Do not connect	☐
GPIO33	I/O	Internal flash - do not use	-	Do not connect	☐
GPIO34	I/O	Spare	-		☐
GPIO35	I/O	PSRAM RESERVED	PSRAM	Internal - do not use	☐
GPIO36	I/O	PSRAM RESERVED	PSRAM	Internal - do not use	☐
GPIO37	I/O	PSRAM RESERVED	PSRAM	Internal - do not use	☐
GPIO38	I/O	Spare	-		☐
GPIO39	I/O	I2C SCL (Wire0)	I2C	DS3231 + ABP2 (4.7k pull-up)	☐
GPIO40	INPUT	Encoder CLK (clock)	DIGITAL	Interrupt driven	☐
GPIO41	I/O	Touch TP_SCL (Wire1 / I2C1)	I2C	GT911 capacitive touch controller	☐
GPIO42	INPUT	Touch TP_INT (interrupt)	DIGITAL	GT911 interrupt - INT floats HIGH at power-up	☐
GPIO43	OUTPUT	UART0 TX	UART	USB serial / programming	☐
GPIO44	INPUT	UART0 RX	UART	USB serial / programming	☐
GPIO45	I/O	Strapping pin	BOOT	Avoid if possible	☐
GPIO46	INPUT	Strapping pin (input only)	BOOT	Avoid - input only	☐
GPIO47	I/O	Touch TP_SDA (Wire1 / I2C1)	I2C	GT911 capacitive touch controller	☐
GPIO48	OUTPUT	TFT BL (backlight PWM)	PWM	PWM brightness 0-255	☐

I2C Bus 0 - Wire0 (GPIO18 SDA / GPIO39 SCL) - DS3231 RTC + Honeywell ABP2

Device	I2C Address	Power	Pull-ups	Notes
DS3231 RTC module	0x68	3.3V	4.7k on bus	Remove charge resistor if using CR2032
Honeywell ABP2 pressure	0x28	3.3V	4.7k on bus	0-4 bar gauge, barbed port to fuel line

I2C Bus 1 - Wire1 (GPIO47 SDA / GPIO41 SCL) - GT911 Capacitive Touch [NEW]

Device	I2C Address	Power	Pull-ups	Notes
GT911 touch controller	0x5D	3.3V	Internal on FPC	INT floats HIGH on power-up -> default addr 0x5D. No pull-up

Poll every 50ms unconditionally. Status register 0x814E. Must clear buffer flag on BOTH touch-down AND lift events.

SPI2 TFT Display - ST7796S 3.5" IPS 480x320 (GPIO11-15, GPIO48)

TFT Pin	GPIO	Signal	Notes	Checked
GND	--	GND	Common ground	☐
VCC	--	3.3V	From ESP32 3V3 pin	☐
SCL	GPIO11	SPI SCK	SPI2 clock	☐
SDA	GPIO12	SPI MOSI	SPI2 data	☐
RES	GPIO15	Reset	Active LOW	☐
DC	GPIO14	Data/Command	HIGH=data LOW=cmd	☐
CS	GPIO13	Chip select	Active LOW	☐
BL	GPIO48	Backlight	PWM brightness 0-255	☐

Library: TFT_eSPI (Bodmer) | Driver: ST7796_DRIVER | USE_FSPI_PORT=1 | setRotation(1) landscape | SPI_FREQUENCY=20MHz

Capacitive Touch Controller - GT911 (Wire1: GPIO47 SDA / GPIO41 SCL / GPIO42 INT) [NEW]

Signal	GPIO	Direction	Notes	Checked
TP_SDA	GPIO47	I/O	Wire1 I2C data. I2C address 0x5D (INT pin floats HIGH at power-up)	☐
TP_SCL	GPIO41	OUTPUT	Wire1 I2C clock 400kHz	☐
TP_INT	GPIO42	INPUT	Interrupt output. Stays LOW after init without RST. Poll unconditionally.	☐
TP_RST	--	--	Not connected. GT911 uses default address 0x5D.	☐
VCC	--	--	3.3V supplied via display FPC connector	☐
GND	--	--	Common ground	☐

Rotary Encoder KY-040 style (GPIO16, GPIO21, GPIO40)

Encoder Pin	GPIO	Function	Notes	Checked
GND	--	GND	Common ground	☐
+	--	3.3V NOT 5V	Power from 3.3V only - avoids 5V on GPIO	☐
SW	GPIO16	Pololu A via D2 1N4148	Via 1N4148 OR gate - GND toggles power	☐
DT	GPIO21	Encoder direction	Read after CLK interrupt	☐
CLK	GPIO40	Encoder clock	Interrupt driven, debounce 5ms	☐

BTS7960 43A H-Bridge Motor Driver - CN10 6-pin Dupont (off-board on heatsink)

BTS7960 Pin	GPIO / Rail	Function	Notes	Checked
RPWM	GPIO2	Fill direction PWM	0-255 PWM	☐
R_EN	GPIO8	Right enable	Hold HIGH in firmware	☐
LPWM	GPIO9	Drain direction PWM	0-255 PWM	☐
L_EN	GPIO10	Left enable	Hold HIGH in firmware	☐
VCC	5V rail	Logic power	From 5V regulator	☐
GND	GND rail	Ground	Common ground	☐
B+	VBAT+	Motor power +	20A inline fuse mandatory	☐
B-	GND rail	Motor power -	Common ground	☐
M+	Motor	Motor output +	2-pin bullet connector CN7	☐
M-	Motor	Motor output -	2-pin bullet connector CN7	☐

Pololu SV Mini Power Switch

Pololu Pin	Connected to	Notes	Checked
VIN	VBAT+ (direct)	Built-in reverse voltage protection	☐
GND	GND rail	Common ground	☐
VOUT	MP1584EN VIN	Switched VBAT to 5V regulator	☐

A pin	D2 + D3 OR junction	Button SW + GPIO16 via 1N4148 - GND toggles power	☐
OFF pin	GPIO17	Pulse HIGH from ESP32 = forced shutdown	☐
B pin	Not connected	Leave unconnected	☐
GND (SW)	Button SW terminal 2	Button other terminal to GND	☐

Resistor Checklist

Ref	Value	Location	Purpose	Fitted
R1	7.5k	Fill flow SIG to GPIO4	Signal divider series	☐
R2	15k	GPIO4 junction to GND	Signal divider shunt	☐
R3	7.5k	Drain flow SIG to GPIO5	Signal divider series	☐
R4	15k	GPIO5 junction to GND	Signal divider shunt	☐
R5	7.5k	Tank full SIG to GPIO6	Signal divider series	☐
R6	15k	GPIO6 junction to GND	Signal divider shunt	☐
R7	100k	VBAT+ to GPIO1	Battery divider series	☐
R8	33k	GPIO1 junction to GND	Battery divider shunt	☐
R9	4.7k	GPIO18 SDA to 3.3V	I2C0 SDA pull-up	☐
R10	4.7k	GPIO39 SCL to 3.3V	I2C0 SCL pull-up	☐
R11	15k	GPIO7 to 3.3V	Tank detect pull-up	☐
R12	100k	Pololu /OFF to GND (optional)	GPIO17 pull-down	☐

Note: GT911 touch uses internal pull-ups on the display FPC - no external pull-ups needed on Wire1 (GPIO41/GPIO47).

Diode Checklist

Ref	Type	Anode	Cathode	Purpose	Fitted
D2	1N4148	Encoder SW output	Pololu A pin	OR gate - button side	☐
D3	1N4148	GPIO16	Pololu A pin	OR gate - GPIO side	☐

Power Supply Checklist

Item	Specification	Check before power-on	Done
5V regulator output	Set to 5.00V exactly	Measure with multimeter BEFORE connecting to board	☐
VBAT+ rail	9.0 - 12.6V	Correct polarity at XT60	☐
5V rail	5.0V +/-0.1V	Measure at ESP32 5V pin	☐
3.3V rail	3.3V +/-0.1V	Measure at ESP32 3V3 pin	☐
GND continuity	All GND points common	Check with continuity beeper	☐
20A fuse	Inline on VBAT+ to BTS	Fitted before motor testing	☐
BTS7960 heatsink	Mounted off-board	Aluminium bracket on case wall	☐

Button: SW+D2->Pololu A (GND toggles) | GPIO16+D3->A | Short press=firmware func | Long 3s+=GPIO17 HIGH=OFF | Rev 3.0